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Metabolism of Benzoic Acid by Bacteria: 3,5-Cyclohexadiene-1,2-Diol-1-Carboxylic Acid Is an Intermediate in the Formation of Catechol

ALBEY M. REINER

Department of Bacteriology and Immunology, University of California, Berkeley, California 94720, and the Department of Microbiology, University of Massachusetts, Amherst, Massachusetts 01002

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3,5-Cyclohexadiene-1,2-diol-1-carboxylic acid (1,2-dihydro-1,2-dihydroxybenzoic acid) is converted enzymatically to catechol in cell extracts from *Acinetobacter*, *Alcaligenes*, *Azotobacter*, and three *Pseudomonas* species. This enzymatic activity is present only in cultures which have been grown in the presence of benzoic acid, and which convert benzoic acid to catechol rather than to protocatechuic acid. The reaction is assayed by the concomitant formation of reduced nicotinamide adenine dinucleotide from nicotinamide adenine dinucleotide. The conversion of [^{14}C]benzoic acid to [^{14}C]dihydrodihydroxybenzoic acid is demonstrated in cell extracts. A scheme for the conversion of benzoic acid to catechol in bacteria is presented, involving the formation of dihydrodihydroxybenzoic acid from benzoic acid by a dioxygenase which is unstable in cell extracts, followed by the dehydrogenation and decarboxylation of dihydrodihydroxybenzoic acid to catechol by a previously undescribed enzyme. Experiments with anthranilic acid and phthalic acid suggest that dihydrodihydroxybenzoic acid is a metabolite unique to benzoic acid metabolism. Two new methods for assaying benzoic acid dioxygenase are suggested.

The catabolism of aromatic compounds by microorganisms generally involves an extended series of enzymatic conversions of the aromatic compound to aliphatic intermediates of major metabolic pathways. Several such pathways have been studied in considerable detail (5; S. Dagley, *Advances in Microbial Physiology*, vol. 6, *in press*). Benzoic acid as sole carbon and energy source can support growth of a variety of microorganisms. Its degradation via β-ketoadipic acid and the convergent pathway of tryptophan degradation are shown in Fig. 1. The conversions of benzoic acid and anthranilic acid to catechol are exceptions to an otherwise well characterized scheme. Their oxidation in cell extracts has been demonstrable only at low levels. Intermediates in the conversion of these compounds to catechol have been suggested, but no evidence of intermediates has been obtained previously (11, 22). Recently a mutant strain of *Alcaligenes eutrophus* was shown to accumulate from benzoic acid a previously unknown compound (18), 1,2-dihydro-1,2-dihydroxybenzoic acid (DHB; see Fig. 2). (The systematic name is 3,5-cyclohexadiene-1,2-diol-1-carboxylic acid.) Evidence presented here shows that this compound is a general intermediate in the conversion of benzoic acid to catechol by bacteria.

MATERIALS AND METHODS

Source and growth of bacteria. *A. eutrophus* strain B9 (13, 18) was supplied by B. Johnson. The following strains were from the culture collection of the Department of Bacteriology and Immunology, University of California, Berkeley: *Acinetobacter calcoaceticus* 73 (formerly designated *Moraxella calcoacetica* 73; see reference 3), *Alcaligenes eutrophus* 335 (ATCC 17697, formerly designated *Hydrogenomonas eutropha;* see reference 4), *Azotobacter vinelandii*, *Pseudomonas aeruginosa* 132 (ATCC 17504), *P. cepacia* 249 and *P. cepacia* 382 (ATCC 17616 and ATCC 17759, formerly designated *P. multivorans* 249 and 382; see reference 2), *P. putida* 90 (ATCC 12633), and *P. testosteroni* 16 (ATCC 15668). Cells were grown in liquid HMB medium (20) at 30 C on a rotary shaker. Carbon sources other than succinic acid were supplied so that their concentrations in the culture medium never exceeded 2 mM. Succinic acid, if present, was supplied at 7 mM.

Preparation of cell extracts. Cultures were harvested by centrifugation at 6×10^8 to 8×10^8 cells/ml, determined turbidimetrically. Cells were washed and suspended at 10^{11}/ml in 0.02 M tris(hydroxymethyl)aminomethane (Tris)-hydrochloride (*p*H 8.0) and were disrupted by sonic oscillation for three periods of 2 min

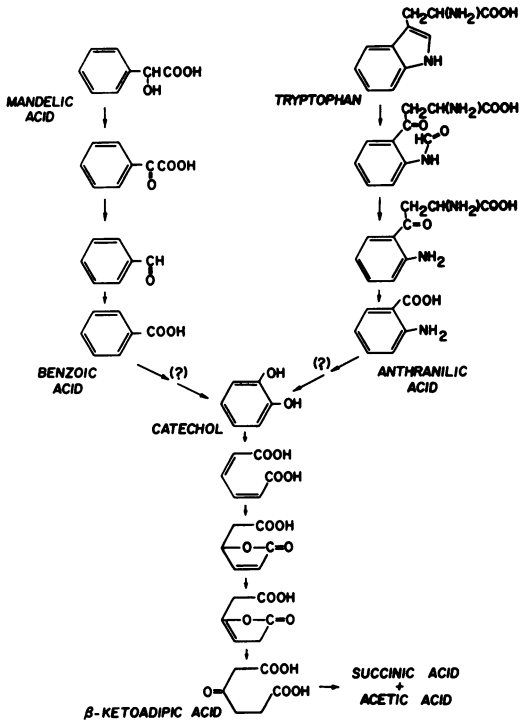


FIG. 1. *β-Ketoadipic acid pathway often used by bacteria for the degradations of benzoic acid and tryptophan.*

each. Debris was removed by centrifugation at 15,000 × *g* for 15 min. Supernatant fractions of extracts were prepared by centrifugation at 100,000 × *g* for 90 min.

Assay for DHB dehydrogenase. Assays were performed at room temperature by adding 2 to 20 μg of cell protein (supernatant fraction) to a 1-ml cuvette containing 20 μmoles of Tris-hydrochloride (*p*H 8.0), 0.5 μmole of nicotinamide adenine dinucleotide (NAD), and 1 μmole of DHB. The increase in absorption at 340 nm was recorded. No measurable increase was observed in the absence of DHB under these conditions. For all extracts except that of *P. aeruginosa*, activity was constant for several minutes. For *P. aeruginosa*, activity declined rapidly and was measured over the interval from 15 to 30 sec after enzyme addition (several measurements averaged). One unit of activity is defined as 1 μmole of reduced NAD (NADH) formed per min.

Reaction products from [^{14}C] benzoic acid. Each reaction took place in a 25-ml flask sealed by a serum cap. Through the cap hung a plastic center well containing 0.15 ml of 1 M Hyamine to trap CO_2. The flask contained 0.5 to 1 mg of cell protein (supernatant fraction) from extracts of cells grown on benzoic acid. The reaction was started by injecting, through the serum cap, 75 μmoles of Tris-hydrochloride, (*p*H 7.0), 0.5 μmole of NADH, 0.05 μmole of $FeSO_4$, 0.1 μmole of benzoic acid, and [*carboxyl*-^{14}*C*] benzoic acid to reach a final volume of 0.5 ml. Incubation was at 25 C with gentle shaking. The reaction was stopped after 4 min by injecting 0.2 ml of a 1/100th dilution of glacial acetic acid. This rendered the medium sufficiently acidic to release CO_2 from solution but not to decompose the acid-labile DHB. The flask was gently shaken for an additional 15 min, and the center well was then removed and placed in a vial containing Bray's scintillation fluid for measurement of its [^{14}C]CO_2 content. Other reaction products were separated by thin-layer chromatography.

Chromatography of ^{14}C reaction products. The acidified contents of reaction flasks were centrifuged to remove the precipitates, and 2-μliter samples of each supernatant were spotted onto three silica gel, thin-layer chromatography sheets (Eastman 6060, type K301R with fluorescent indicator). Unlabeled benzoic acid, salicylic acid, and DHB were added to each spot as markers. Each sheet was then developed in one of three solvent systems and dried. The marker compounds, which were visible as quenching spots, were outlined under ultraviolet light. Sheets were cut into strips corresponding to each spot originally applied, and each strip was cut along its length into 20 to 30 segments. The radioactivity of each segment was measured by scintillation. Solvent A consisted of benzene-methanol-acetic acid (8:3:1); benzoic acid, R_F 0.67; DHB, R_F 0.23. Solvent B consisted of benzene-methanol-acetic acid (4:6:1); benzoic acid, R_F 0.79; DHB, R_F 0.64. Solvent C consisted of ether-toluene-acetic acid (10:2:1); benzoic acid, R_F 0.62; DHB, R_F 0.19.

Other methods. Protein concentrations were measured by the method of Lowry et al. (16), with crystalline bovine serum albumin as standard. Catechol was measured by the method of Arnow (1). Respiration rates were measured with a Gilson Differential Respirometer.

Materials. [*Carboxyl*-^{14}*C*]benzoic acid (56 μCi/μmole) was purchased from Amersham-Searle Corp., Arlington Hts., Ill., and was purified by thin-layer chromatography (silica gel, solvent A). DHB (sodium salt) was prepared biologically as described previously (18). Other chemicals were the highest qualities available from commercial sources and were used as obtained.

RESULTS

Conversion of DHB to catechol in cell extracts. An NAD-dependent DHB dehydrogenase activity is found at high levels in extracts prepared from bacteria grown in the presence of benzoic acid. Approximately 1 mole of catechol and 1 mole of NADH are formed from 1 mole of DHB and 1 mole of NAD, and neither product is formed without the addition of both DHB and NAD (Table 1). This stoichiometry is demonstrable by centrifuging extracts to remove most of the NADH oxidase with the pellet fraction, and heating to inactivate the catechol oxygenase present.

The measured levels of enzymatic activity, here called DHB dehydrogenase, are shown in Table 2 for a number of species which can grow at the expense of benzoic acid. The activities in extracts of induced cells are sufficient to account

TABLE 1. *Stoichiometry of the DHB dehydrogenase reaction*[a]

DHB added (mM)	NAD added (mM)	Catechol formed (mM)	NADH formed (mM)
0	1.0	0.000	0.000
0.1	1.0	0.099	0.084
0.2	1.0	0.196	0.161
0.3	1.0	0.293	0.253
1.0	0	0.001	0.002
1.0	0.1	0.092	0.090
1.0	0.2	0.197	0.177
1.0	0.3	0.310	0.276

[a] Assays were performed by adding to a 1-ml cuvette containing 20 mM Tris-hydrochloride (*p*H 8.0) and reactants, as shown, 40 μg of protein from an extract of *Alcaligenes eutrophus* 335 (supernatant fraction) which had been heated for 15 min at 49 C. Absorption at 340 nm was recorded. When no further increase in absorption occurred, the catechol concentration was determined by the method of Arnow (1). NADH concentration is based on $\mathcal{E}$ = 6,220.

for the rates of consumption of benzoic acid by intact cells, measured separately by respirometry. No activity was detected in cells which were not induced by benzoic acid. Properties of purified DHB dehydrogenase [3,5-cyclohexadiene-1,2-diol-1-carboxylic acid (NAD) oxidoreductase (decarboxylating)] and details of the catalyzed reaction will be presented separately.

The physiological role of DHB dehydrogenase is supported by its absence from two strains which were included as controls (Table 2). *A. eutrophus* strain B9 is a mutant which is unable to convert benzoic acid to catechol and which accumulates DHB (18). My results show that its block is due to the absence of DHB dehydrogenase. *P. testosteroni* metabolizes benzoic acid via *m*-hydroxybenzoic acid and protocatechuic acid rather than via catechol (23). The absence of activity in the *P. testosteroni* extract indicates that the DHB dehydrogenase is specific for metabolism of benzoic acid via catechol.

Conversion of benzoic acid to DHB in extracts. Because of the instability of benzoic acid oxygenase in cell extracts, high concentrations of cell protein were required to demonstrate benzoic acid oxidation in extracts, in confirmation of previous reports (11). Since the conversion of DHB to catechol proceeds rapidly in extracts, precautions were necessary to show the formation of DHB from benzoic acid. This was accomplished by choosing conditions relatively unfavorable for DHB dehydrogenase and by using isotope labeling.

Cell extracts were incubated with [*carboxyl*-^{14}C]benzoic acid. At the buffer conditions selected (0.15 M Tris, *p*H 7.0), DHB dehydrogenase activity was more than 90% inhibited, whereas benzoic acid oxidation was near its maximum rate. The reaction products were separated, with results as shown in Table 3. For each extract tested, a portion of the label was recovered as [^{14}C]DHB. When unlabeled DHB was added to the reaction mixture, additional label was "trapped" as DHB, with a corresponding reduction in [^{14}C]CO_2 release. The identity of the [^{14}C]DHB was confirmed by cochromatography with unlabeled DHB in three solvent systems. No radioactive spots other than those of [^{14}C]DHB and [^{14}C]benzoic acid were detected.

In the formation of DHB from benzoic acid, an electron donor is needed (Fig. 2). That NADH is required and that reduced nicotinamide adenine dinucleotide phosphate (NADPH) cannot substitute was shown for *A. calcoaceticus* (Table 4). DHB dehydrogenase, which was used in this assay to catalyze the release of [^{14}C]CO_2 from [^{14}C]DHB, was present in excess. The

TABLE 2. *DHB dehydrogenase activity in supernatant fractions of cell extracts*

Bacterial strain	Specific activities[a] in extracts from cells grown with			
	Benzoic acid	*p*-OH benzoic acid	Tryptophan	Succinic acid
Acinetobacter calcoaceticus	0.92	<0.005	<0.005	<0.005
Alcaligenes eutrophus 335	0.85	<0.005	<0.005	<0.005
Azotobacter vinelandii	0.36			<0.005
Pseudomonas aeruginosa 132	0.20			<0.005
Pseudomonas cepacia 382	0.79	<0.005		<0.005
Pseudomonas putida 90	0.64	<0.005		<0.005
Alcaligenes eutrophus B9	<0.005			
Pseudomonas testosteroni 16	<0.005			

[a] Expressed as micromoles of NADH formed per minute per milligram of protein.

TABLE 3. *Products recovered after incubation of [carboxyl-^{14}C]benzoate with cell extracts*[a]

Product	Source of extract					
	Alcaligenes eutrophus 335		*Acinetobacter calcoaceticus* 73		*Pseudomonas putida* 90	
	No DHB added	40 mM DHB added	No DHB added	40 mM DHB added	No DHB added	40 mM DHB added
[^{14}C]CO_2	1,370	14	901	25	162	14
[^{14}C]DHB	1,092	2,539	362	1,331	1,161	1,214
[^{14}C]benzoic acid	3,617	3,709	5,144	4,812	4,660	4,710

[a] Results were essentially the same with each of three solvent systems used and are shown here for solvent A. Figures indicate counts per minute per microliter of reaction mixture. At start of reactions, [^{14}C]benzoic acid was present at 6,400 counts per min per μliter.

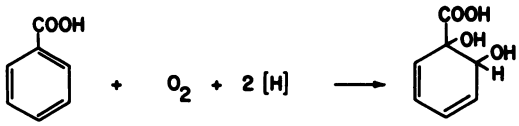


FIG. 2. *Formation of 1,2-dihydro-1,2-dihydroxybenzoic acid from benzoic acid.*

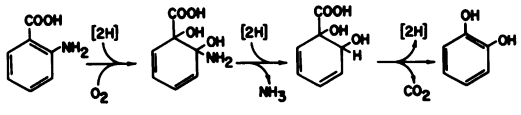


FIG. 3. *Possible route for the conversion of anthranilic acid to catechol via 1,2-dihydro-1,2-dihydroxybenzoic acid.*

TABLE 4. *Pyridine nucleotide requirement for DHB formation in Acinetobacter calcoaceticus*[a]

Cofactor added (each at 1 mM)	[^{14}C]CO_2 released (counts/min)
NADH	12,468
NADPH	33
NAD	768
NAD + NADPH	872
None	60

[a] Reactions were as described for [^{14}C]benzoic acid reaction products (Materials and Methods), except that flasks contained 25 μmoles of Tris-hydrochloride (*p*H 7.0), NADH was present only when shown, and reactions proceeded for 10 min. Initially, 16,000 counts/min of [^{14}C]benzoic acid and 0.5 mg of extract (supernatant fraction) were present.

NAD which it requires is presumably formed from NADH in the first reaction. It was reported previously that benzoic acid oxidation proceeds with either NADH or NADPH in *P. flourescens* (11).

No evidence for DHB in related pathways. Experiments were performed to investigate whether anthranilic acid (*o*-aminobenzoic acid) or phthalic acid (*o*-carboxybenzoic acid) were metabolized via DHB.

The conversion of anthranilic acid to catechol is not well understood (22) and could involve DHB and DHB dehydrogenase, as shown in Fig. 3. To test for this pathway in *Acinetobacter*, cells were induced to metabolize anthranilic acid, and the level of DHB dehydrogenase was measured. Induction was achieved by growing the cells on tryptophan, which is metabolized via anthranilic acid and catechol in many bacteria (*see* Fig. 1 and reference 10). Rapid metabolism of anthranilic acid and catechol by whole cells was confirmed by respirometry. No DHB dehydrogenase was found, however (Table 2). In *A. eutrophus*, mutant strain B9 lacks DHB dehydrogenase (Table 2). The ability of this strain to respire on anthranilic acid (after induction with tryptophan) was found to be unimpaired, indicating that *A. eutrophus* does not metabolize anthranilic acid via DHB.

Four of 267 strains which had been screened for growth on a variety of carbon sources had been found to grow on phthalic acid (20). Two of those strains, *P. cepacia* 249 and *P. testosteroni* 139, were grown on phthalic acid and tested for their ability to respire at the expense of catechol and protocatechuic acid. The results (Table 5) indicate that phthalic acid is metabolized via protocatechuic acid rather than via catechol in these species, in confirmation of an earlier study on a soil pseudomonad isolated by enrichment with phthalic acid (19).

DISCUSSION

A. eutrophus mutant strain B9 recently was shown to convert benzoic acid to DHB, apparently via a peroxidation mechanism (18). Evidence presented here shows that extracts from several bacterial species convert benzoic acid to DHB and convert DHB to catechol with the concommitant reduction of NAD. These results are summarized in the overall scheme shown in Fig. 4. The conversion of benzoic acid to catechol

TABLE 5. *Rates of respiration of phthalic acid-grown cultures*[a]

Bacterial strain	Rate of oxygen consumption with		
	Phthalic acid	Proto-catechuic acid	Catechol
	μl/min/10⁹ cells		
Pseudomonas cepacia 249	4.7	2.7	0.17
Pseudomonas testosteroni 16	4.0	3.1	0.10

[a] Cultures were washed and suspended in 0.05 M sodium-potassium phosphate buffer, *p*H 6.8. Substrates were tipped from side-arm flasks to reach final concentrations of 1 mM. O_2 consumption was measured for 20 min at 30 C in a Gilson Differential Respirometer.

thus involves the formation of DHB by the benzoic acid oxygenase system which is unstable in extracts, and the dehydrogenation and decarboxylation of DHB by a previously unknown enzyme which is stable in extracts. This scheme is analogous to that presented for the oxidation of benzene to catechol in *P. putida* (6, 7). Evidence that a single enzyme catalyzes the oxidative decarboxylation of DHB to catechol will be presented in a study of DHB dehydrogenase (A. M. Reiner, *manuscript in preparation*).

DHB is implicated as an intermediate between benzoic acid and catechol in the four bacterial genera tested here and presumably is an intermediate generally in the conversion of benzoic acid to catechol in bacteria. However, as decarboxylation can occur also at unhydroxylated ring positions (8, 19), the conversion of benzoic acid to catechol via 2,3- or 3,4-dihydroxybenzoic acid and not via DHB may prove to be an alternative route in some other microorganism. Assaying for DHB dehydrogenase provides an easy means for screening organisms according to their early steps in benzoic acid oxidation. Two other routes of benzoic acid oxidation not involving catechol are already known (12, 23).

Three natural *ortho*-substituted benzoic acids can support growth of certain bacteria. Results presented here indicate that DHB is not an intermediate in the metabolism of anthranilic acid and phthalic acid. The conversion of salicylic acid (*o*-hydroxybenzoic acid) to catechol has been studied in detail elsewhere and is known not to involve an intermediate (14, 21, 24). Since *meta*- and *para*-substituted benzoic acids are not generally metabolized via catechol, it appears that DHB is a metabolite unique to the benzoic acid pathway. For the benzoic acid and *p*-OH benzoic acid pathways, cross-induction of certain enzymes of one pathway by metabolites of the other has been observed (10, 15, 17). This was not found to occur with DHB dehydrogenase, and there is no indication that this enzyme is induced under any conditions except during growth on benzoic acid.

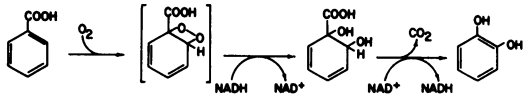


FIG. 4. *Conversion of benzoic acid to catechol via 1,2-dihydro-1,2-dihydroxybenzoic acid in bacteria.*

DHB does not support growth of any of the bacteria used in this study. This is not unexpected for an intermediary metabolite whose polarity would make passive transport through the cell membrane difficult and which is unlikely to be found extracellularly in sufficient amounts to warrant an active transport system. Mutants have been isolated which are able to grow slowly at the expense of DHB.

The benzoic acid oxygenase system previously had been assayed by oxygen consumption (11). Here an alternative assay is used which measures the conversion of [*carboxyl*-^{14}*C*]benzoic acid to [^{14}C]DHB in the presence of an excess of DHB dehydrogenase to catalyze the release of [^{14}C]CO_2. A simpler assay is possible in *A. eutrophus* B9 extracts. NADH which is oxidized in the oxygenase reaction is normally re-formed by DHB dehydrogenase in the next reaction step. In B9, however, the dehydrogenase step is blocked and a net benzoic acid-dependent oxidation of NADH can be measured in extracts. This assay could facilitate a future study of the benzoic acid oxygenase system.

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